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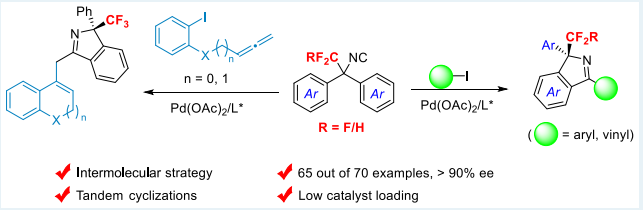
Enantioselective Construction of 1*H*-Isoindoles Containing Tri- and Difluoromethylated Quaternary Stereogenic Centers via Palladium-Catalyzed C−H Bond Imidoylation

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ABSTRACT: An intermolecular enantioselective synthesis of 1*H*-isoindoles containing tri- and difluoromethylated quaternary stereogenic centers through a palladium-catalyzed desymmetric C−H bond imidoylation has been developed. α,α-Diaryl tri- and difluoroethylated isocyanides acted as powerful precursors of chiral 1*H*-isoindoles, in which the fluoroalkyl group was proven to play a crucial role in both the stereochemistry and reaction efficiency. In addition, an allene insertion cascade was realized, offering rapid access to diverse C1-tethered bis-heterocyclic scaffolds with good yields and high enantioselectivities. The reactions proceeded smoothly under mild conditions with low catalyst loadings.



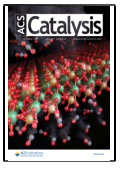
KEYWORDS: *tri- and difluoromethylated quaternary stereogenic center, 1H-isoindole, palladium, imidoylation, enantioselective C−H activation*

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Organic compounds incorporated with fluorine atoms have witnessed broad applications in agro- and medicinal chemistry.[1] The solubility, lipophilicity, binding affinity, and metabolic stability of drug candidates can be significantly altered due to the unique electronic and steric properties of fluoroalkyl moieties. A difluoromethyl group (CF_2H) with an acidic C−H bond can play the role of a lipophilic hydrogen-bond donor and a bioisostere of alcohols and thiols in drug design.[2] In addition to fluoroalkylarenes,[3] tri- and difluoromethylated stereogenic centers frequently occur in pharmaceuticals and lead compounds (Figure 1a) and the catalytic construction of such stereocenters has attracted much attention from the synthetic community.[4] In this regard, asymmetric functionalization of prochiral fluoroalkylated compounds is one of the most important categories.[5] For instance, the use of catalytic enantioselective addition reactions of fluoroalkyl ketones,[6] ketoimines,[7] and Michael acceptors[8] was proven to be a powerful strategy for the synthesis of enantioenriched fluoroalkylated alcohols and amines. Another well-established approach is the transformation of α-trifluoromethyl carbanions, which have traditionally suffered from the side reaction of β-fluoro elimination.[9] Overall, the reported successful strategies[10] mostly relied on transformations of substrates containing unsaturated double bonds and were mainly limited to trifluoromethyl compounds. The construction of difluoromethylated stereogenic centers has been far less investigated, and only a few examples have been reported.[11] Furthermore, the synthesis of complex molecules with switchable CF_3- or CF_2H-containing stereogenic centers by using the same synthetic method has been even more rarely developed.[12]

The isoindole core is ubiquitous in bioactive molecules and natural products.[13] Although a variety of catalytic enantioselective methods have been developed to access isoindolines[14−18] and isoindolones[19−22] (reduced and oxidized isoindoles) by the groups of Enders,[14] Sasai,[15] Gong,[16] Jarvo,[17] Wang,[18] Matsunaga,[19] Cramer,[20] Nishimura,[21] and Wang[22] and others, the synthesis of 1*H*-isoindoles has mostly been confined to racemic versions.[23] In 2018, Cramer and coworkers developed the first Pd(0)-catalyzed intramolecular C−H bond imidoylation reaction to afford 1*H*-isoindoles with quaternary stereogenic centers (Figure 1b).[24] The enantioselective C−H bond activation[25] strategy utilized trifluoromethyl-substituted imidoyl chlorides as internal electrophiles. Herein, we report an intermolecular protocol for the construction of 1*H*-isoindoles bearing both tri- and difluoromethylated quaternary stereogenic centers through palladium-catalyzed C−H bond imidoylation.[26] α,α-Diaryl tri- and difluoroethylated isocyanides were utilized as coupling partners of aryl and alkenyl iodides, offering access to diverse 1*H*-

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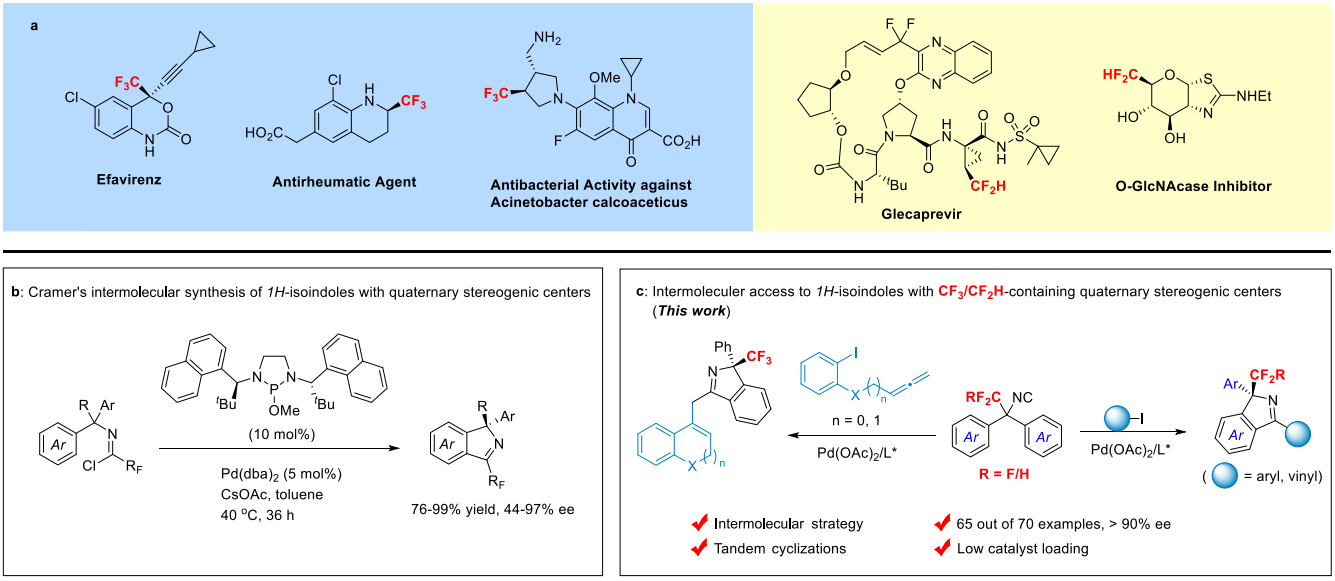


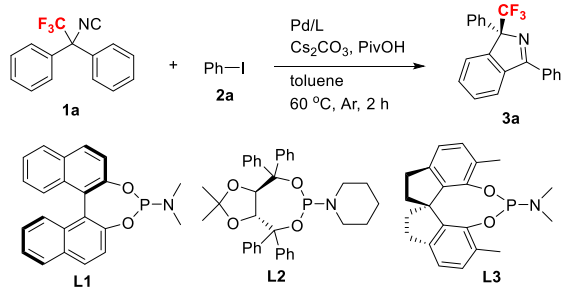
Figure 1. (a) Selected drugs and bioactive molecules. (b) Intramolecular synthesis of 1*H*-isoindoles with quaternary stereogenic centers. (c) Intermolecular synthesis of 1*H*-isoindoles bearing tri- and difluoromethylated quaternary stereogenic centers.

isoindoles in high yields and enantioselectivities (up to 99% yield and 99% ee) (Figure 1c). Furthermore, several kinds of C1-tethered bis-heterocyclic scaffolds were enantioselectively accessed through an allene insertion cascade. The reactions proceeded smoothly under mild conditions with low catalyst loadings. Control experiments revealed that the fluoroalkyl group might play a crucial role in the stereochemistry and reaction efficiency.

The study was initiated by employing (2,2,2-trifluoro-1-isocyanoethane-1,1-diyl)dibenzene (**1a**) and iodobenzene (**2a**) as model substrates. When **1a** was added dropwise via a syringe pump over 2 h, the racemic product **3a** was generated in 97% yield under the catalysis of $Pd(OAc)_2$ (10 mol %) and PPh_3 (20 mol %) with Cs_2CO_3 (1.0 equiv) and PivOH (0.6 equiv) as additives (Table 1, entry 1). Next, chiral phosphoramidite ligands L1−L3 were tested for asymmetric catalysis (entries 2−4), and product **3a** was formed in 94% yield and 98% ee with the SPINOL-derived phosphoramidite ligand L3 (entry 4). When bidentate BINAP was used as the ligand, both the efficiency and enantioselectivity of the reaction obviously decreased (entry 5). Other palladium catalysts such as $PdCl_2$ and $Pd_2(dba)_3$ resulted in lower yields of 60% and 49%, respectively (entries 6 and 7). Satisfyingly, when 1 mol % of catalyst was used, the product was delivered in 93% yield and 97% ee (entry 9). The enantioselectivity decreased to 89% by using bromobenzene as an electrophilic coupling partner of **1a** (entry 10).

With the optimized conditions in hand, the scope for the construction of 1*H*-isoindoles containing trifluoromethylated stereogenic centers was first investigated (Scheme 1). Screening of functional groups on the aryl iodides revealed that both electron-donating (Me, ^{t}Bu, and OMe) and electron-withdrawing (F, Cl, Br, CF_3, CO_2Me, CHO, CN, and NO_2) groups were well tolerated. The corresponding products **3a−u** were generated in high yields and enantioselectivities. When the reaction was conducted with 1-iodo-2-methylbenzene as the coupling partner, 2-methylated product **3b** was obtained in 90% yield and 89% ee, suggesting that the enantioselectivity was slightly influenced by steric hindrance. Notably, the C−Br bond remained stable under the applied conditions and

Table 1. Optimization of the Conditions[a]



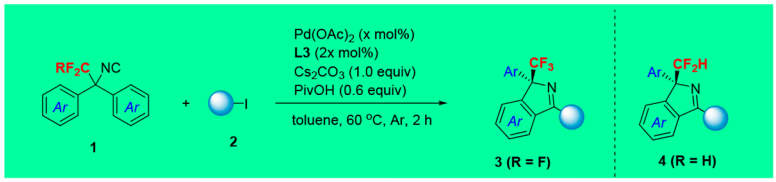
entry	catalyst	ligand	yield (%)	ee (%)
1	$Pd(OAc)_2$	PPh_3	97	
2	$Pd(OAc)_2$	L1	96	92
3	$Pd(OAc)_2$	L2	91	11
4	$Pd(OAc)_2$	L3	94	98
5	$Pd(OAc)_2$	BINAP	30	8
6	$PdCl_2$	L3	60	95
7	$Pd_2(dba)_3$	L3	49	99
8	$Pd(OAc)_2$	L3	93	97[b]
9	**$Pd(OAc)_2$**	**L3**	**93**	**97**[c]
10	$Pd(OAc)_2$	L3	92	89[c,d]

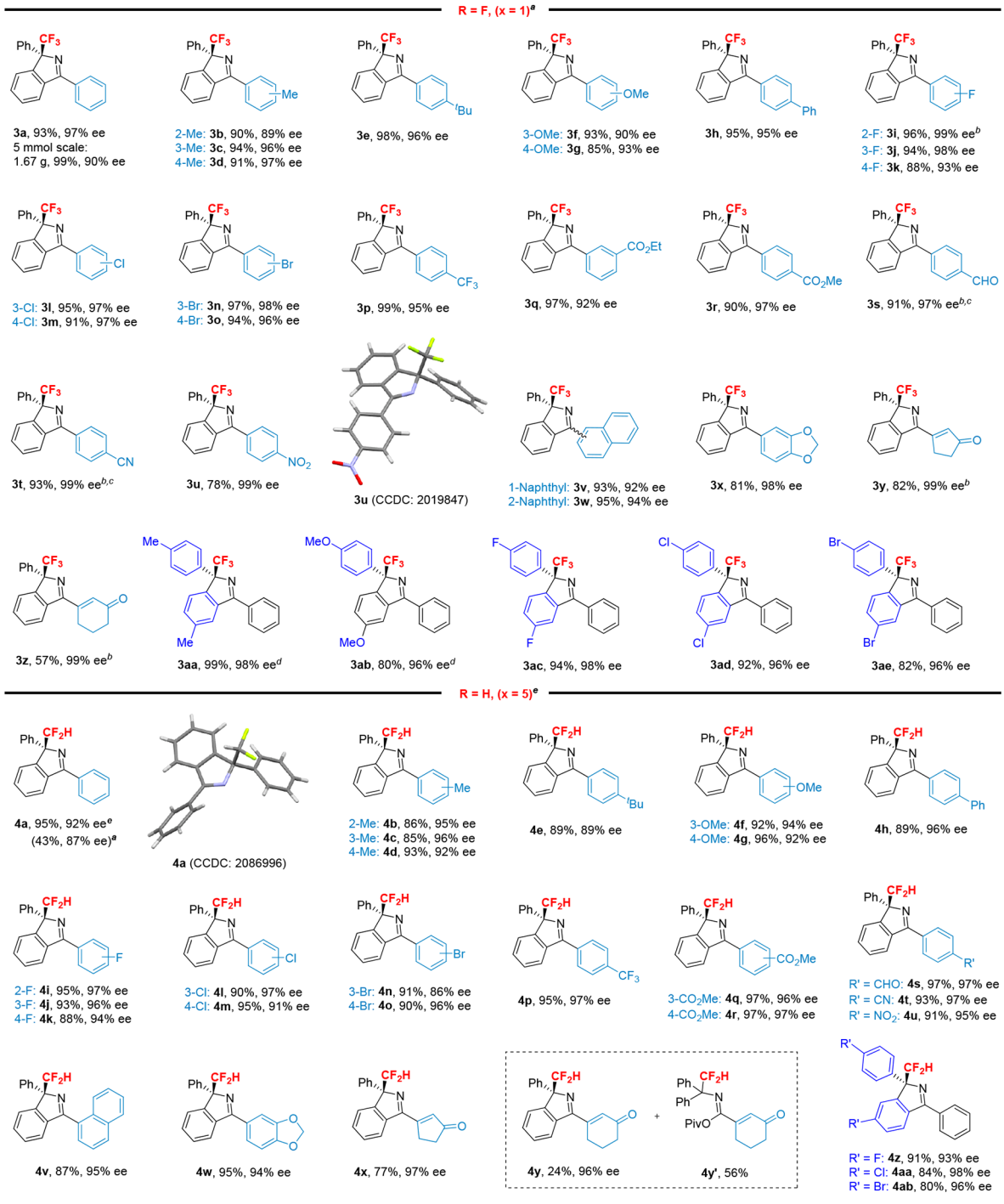
[a]Reaction conditions unless specified otherwise: **1a** (0.1 mmol), **2a** (0.12 mmol), Pd catalyst (10 mol %), ligand (20 mol %), Cs_2CO_3 (0.1 mmol), PivOH (0.06 mmol), 60 °C, under Ar. A solution of **1a** in 1.0 mL of toluene was added to the reaction mixture (in 1.0 mL of toluene) dropwise via a syringe pump in 2 h. Yields of isolated products are given. [b]Pd catalyst (5 mol %), ligand (10 mol %). [c]The reaction was carried out on a 0.5 mmol scale with Pd catalyst (1 mol %) and ligand (2 mol %). [d]Bromobenzene as an electrophile.

bromosubstituted products **3n,o** were obtained in 97% yield with 98% ee and 94% yield with 96% ee, respectively, which could facilitate further transformations. Naphthyl- and heteroaryl-substituted products **3v−x** were also obtained in high yields and ee. In addition, alkenyl iodides could act as efficient electrophilic coupling partners with a slightly improved catalyst loading of 5 mol %, affording alkenylated 1*H*-isoindoles **3y,z** in good yields and high enantioselectivities.

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Scheme 1. Reaction Scope[a]

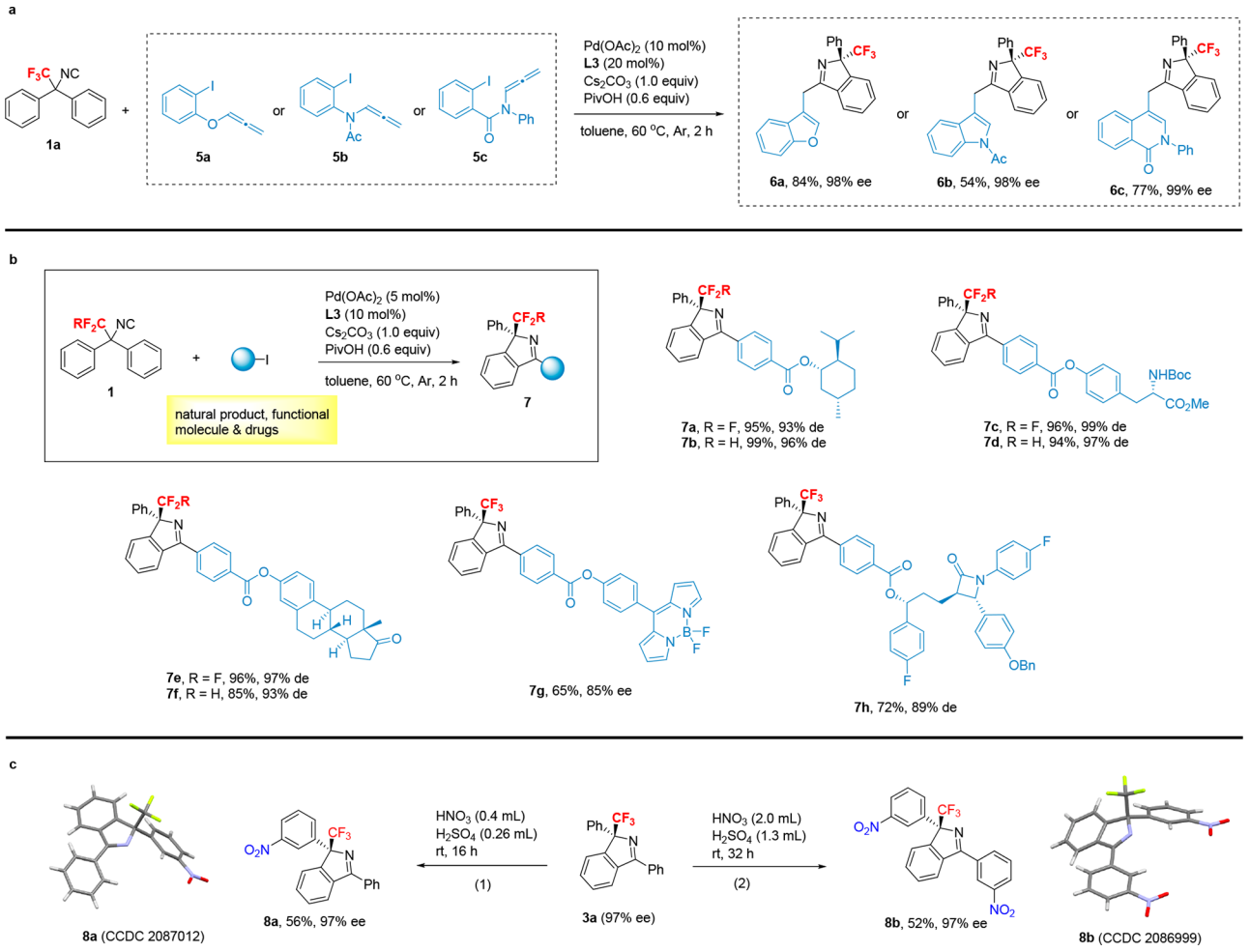




[a]Reaction conditions unless specified otherwise: **1** (0.5 mmol), **2a** (0.75 mmol), $Pd(OAc)_2$ (1 mol %), **L3** (2 mol %), Cs_2CO_3 (0.5 mmol), PivOH (0.3 mmol), 60 °C, under Ar. A solution of **1a** in 3 mL of toluene was added to the reaction mixture (in 3 mL of toluene) via a syringe pump over 2 h. Yields of isolated products are given. [b]$Pd(OAc)_2$ (10 mol %), **L3** (20 mol %). [c]70 °C. [d]$Pd(OAc)_2$ (5 mol %), **L3** (10 mol %). [e]**1** (0.1 mmol), **2a** (0.15 mmol), $Pd(OAc)_2$ (5 mol %), **L3** (10 mol %), Cs_2CO_3 (0.1 mmol), PivOH (0.06 mmol), 60 °C, under Ar. Starting material **1** was added dropwise over 2 h.

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Scheme 2. Tandem Cyclization and Synthetic Applications[a]



[a]Reaction conditions: (a) **1a** (0.1 mmol), **5** (0.15 mmol), $Pd(OAc)_2$ (10 mol %), **L3** (20 mol %), Cs_2CO_3 (0.1 mmol), PivOH (0.06 mmol), 60 °C, under Ar. **1a** was added dropwise over 2 h; (b) $Pd(OAc)_2$ (5 mol %), **L3** (10 mol %); (c) (1) **3a** (0.2 mmol), HNO_3 (0.4 mL), H_2SO_4 (0.26 mL), rt, 16 h, (2) **3a** (0.2 mmol), HNO_3 (2.0 mL), H_2SO_4 (1.3 mL), rt, 32 h. Yields of isolated products are given.

Next, the scope of isocyanides was tested under the standard conditions. Substituents such as Me, OMe, F, Cl, and Br were all well tolerated. Products **3aa**−**ae** were obtained in 80−99% yields and 96−98% ee. A gram-scale preparation of **3a** was conducted without loss of efficiency.

Then our attention switched to the study of construction of CF_2H-containing stereocenters (Scheme 1). When (2,2-difluoro-1-isocyanoethane-1,1-diyl)dibenzene **1** was reacted with iodobenzene **2a** under the standard conditions, the desired product **4a** was generated in only 43% yield and 87% ee, with a large amount of isocyanide remaining unreacted. To our delight, the yield and enantioselectivity were improved to 95% and 92%, respectively, by increasing the catalyst loading to 5 mol %. The efficient synthesis of CF_2H-containing chiral 1*H*-isoindoles was developed by investigating aryl iodides with various functional groups such as Me, ^{t}Bu, OMe, Ph, F, Cl, Br, CF_3, CO_2Me, CHO, CN, and NO_2. Aryl- and heteroarylated 1*H*-isoindoles **4a**−**w** were generated in high yields and ee. Alkenylated products **4x,y** were obtained in 77% yield with 97% ee and 24% yield with 96% ee, respectively. The low yield of **4y** was attributed to the generation of the imidic pivalic anhydride **4y′** as a major byproduct. The isocyanide substrates with F, Cl ,and Br functional groups reacted smoothly, and cyclization products **4z**−**ab** were generated in 80−91% yield and 93−98% ee.

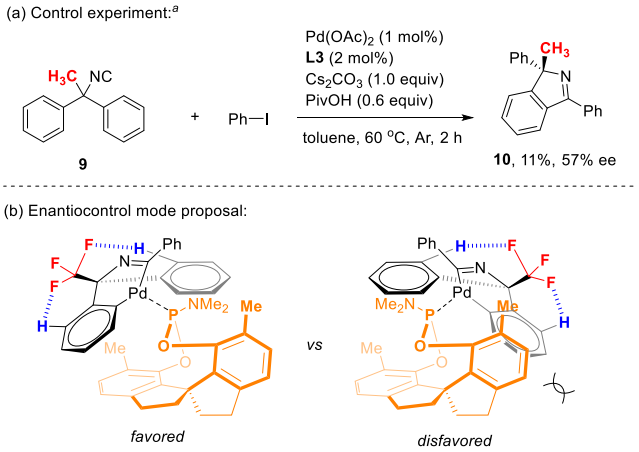
Next, we examined the reactivity of isocyanide insertion into the allyl-palladium intermediate, which was generated by intramolecular allene insertion (Scheme 2a). Satisfyingly, when allene substrates **5a**−**c** were subjected to the reaction with a catalyst loading of 10 mol %, C1-tethered benzofuran/indole/isoquinolinone-isoindole bisheterocyclic scaffolds **6a**−**c** were generated in 54−84% yields and 98−99% ee.

To further demonstrate the utility of this reaction, we tested enantioselective C−H imidoylation with aryl iodides derived from natural products (menthol, tyrosine, and estrone), a dye (BODIPY), and drug molecules (Scheme 2b). The products were uniformly obtained in good yields with high enantiomeric (**7g**; 65% yield and 85% ee) and diastereomeric (**7a**−**f,h**; 72−99% yields and 89−99% de) excesses. Then the transformations of **3a** were carried out as exemplified in Scheme 2c. Selective mono- and dinitration reactions were successfully performed, delivering **8a,b** in 56% and 52% yields, respectively. The chirality remained stable under the applied conditions.

To shed light on the stereochemical outcomes of the cyclization reaction, a control experiment with methylated isocyanide **9** as the substrate was carried out under the standard conditions (Scheme 3a). The imidoylation product

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Scheme 3. Control Experiment and Catalytic Mode Proposal



[a]Reaction conditions: **9** (0.5 mmol), iodobenzene (0.75 mmol), $Pd(OAc)_2$ (1 mol %), **L3** (2 mol %), Cs_2CO_3 (0.5 mmol), PivOH (0.3 mmol), 60 °C, under Ar. Starting material **9** was added dropwise via a syringe pump over 2 h.

10 was obtained in only 11% yield and 57% ee, suggesting that the fluoroalkyl group might play a crucial role in both the enantioselectivity control and the reaction efficiency of the desymmetric C−H imidoylation. On the basis of previous reports,[27] a plausible C−H···F−C hydrogen-bond effect[27c−e] is preliminarily depicted in Scheme 3b.

In summary, we have developed an enantioselective palladium-catalyzed C−H bond imidoylation for the construction of 1*H*-isoindoles containing tri- and difluoromethylated quaternary stereogenic centers with *α*,*α*-diaryl tri- and difluoroethyl isocyanides as coupling partners of aryl and alkenyl iodide derivatives. A variety of chiral 1*H*-isoindoles were synthesized in good yields (up to 99% yield) and enantioselectivities (up to 99% ee). This intermolecular protocol is also compatible with various aryl iodides bearing core structures of natural products, a dye, and drug molecules. In addition, diverse C1-tethered bis-heterocyclic scaffolds were afforded in good yields and high enantioselectivities through an allene insertion cascade. The reactions were carried out under mild conditions with low catalyst loadings. A bioactive evaluation of the chiral products is currently underway in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.1c03682.

Experimental details, characterization data, and X-ray crystallographic data (PDF)

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Notes

The authors declare no competing financial interest.

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